

FormosaNLU-Synth: Filtered Synthetic Data Distillation for Traditional Chinese (Taiwan) Natural Language Understanding

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Abstract

We study whether locally generated and aggressively filtered synthetic data can improve low-resource joint intent classification and slot filling in Traditional Chinese as used in Taiwan. Starting from a 20-shot-per-intent subset of MASSIVE **zh-TW**, a local open-weight teacher generates four kinds of examples and a fixed seven-stage pipeline filters them. The public corpus contains 3,754 released rows. Under a frozen 500-step QLoRA contract, adding filtered synthetic data to the real-only baseline improves three-seed Gemma intent accuracy and exact match by +4.14 and +3.86 percentage points, respectively. A preregistered replication with Phi-4-mini yields corresponding gains of +5.09 and +4.71 percentage points. A single-seed equal-size recipe ablation finds that no leave-one-recipe-out exact-match difference reaches the preregistered 2.5-point detectability threshold. We release the corpus, one Gemma adapter, one Phi adapter, code, machine-readable reports, checksums, and resource accounting. This document is an evidence-bounded technical report rather than a peer-reviewed publication.

1 Introduction

Natural language understanding (NLU) systems for intent classification and slot filling require examples that cover both label semantics and local linguistic variation. This requirement is difficult in a low-resource locale such as Traditional Chinese as used in Taiwan (**zh-TW**). MASSIVE provides a parallel multilingual benchmark with intent and slot annotations (FitzGerald et al., 2023), but a deployment team may still have only a small labeled subset available for adaptation.

FormosaNLU-Synth asks a deliberately narrow question: under one frozen low- resource MASSIVE **zh-TW** contract, can synthetic examples produced by a local open-weight teacher improve a smaller student over the same real-only baseline? The project separates generation, filtering, training, and evaluation; keeps the test split outside every construction decision; and preserves negative results rather than adjusting thresholds after inspection.

The evidence supports three bounded conclusions. First, filtered synthetic data improves the preregistered paired comparison for two student model families. Second, the result survives deterministic robustness probes, although uncertainty is substantial with three seeds. Third, a single-seed equal-size composition study does not isolate a detectable independent contribution from any one recipe. These findings do not establish generalization to other datasets, languages, model families, or training budgets.

2 Related Work

MASSIVE contains one million virtual-assistant utterances across 51 languages, 18 domains, 60 intents, and 55 slot types (FitzGerald et al., 2023). The present work uses its Traditional Chinese (Taiwan) partition and retains its joint intent and slot structure.

Synthetic instruction-data pipelines can bootstrap supervision from model generations and then reject invalid or overly similar examples. Self-Instruct demonstrated this generate–filter–fine-tune pattern for instruction following (Wang et al., 2023). FormosaNLU-Synth applies the general pattern to a fixed structured-prediction schema rather than claiming a new general-purpose instruction-generation algorithm.

The student models are adapted with low-rank updates. LoRA freezes pretrained weights and learns low-rank matrices (Hu et al., 2022); QLoRA extends this idea to backpropagation through quantized base weights (Detmeters et al., 2023). The primary student is the Apache-2.0 Gemma 4 E4B instruction model (Google, 2026). The replication student is the MIT-licensed Phi-4-mini instruction model (Microsoft, 2025).

3 Method

Real-data budget. For each MASSIVE intent, the low-resource training set contains at most 20 real examples selected with seed 42, producing 1,176 real examples. Validation and test rows never enter generation or training-data assembly.

Local generation. The teacher is `qwen3.6:27b`, served locally through Ollama. It produces four recipe families: label-preserving paraphrases, slot substitution followed by constrained rewriting, locale-aware noise and code-switching, and hard negatives between confusable intents. Labels are constructed by the pipeline; there is no row-by-row human annotation. The teacher, primary student, and independent judge belong to different model families.

Fixed filtering. The pipeline applies schema validation (F1), label-contract checks (F2), grounded slot checks (F3), Taiwan locale and language checks (F4), seed-copy, duplicate, and outlier removal (F5), and validation/test contamination exclusion (F6). These fixed stages reduce 11,264 generated rows to 3,760. A stratified F7 audit with the independent `gpt-oss:20b` judge rejects six rows from the public release, leaving 3,754 rows. The already frozen training experiments retain the 3,760-row F1–F6 corpus; F7 is a release-only exclusion and does not retroactively change the experimental contract.

The largest removal is synthetic near-duplication: 4,596 rows are rejected as near duplicates. The preregistered target of 8,000–9,000 accepted examples is therefore missed. Thresholds were not relaxed after this shortfall was observed.

Structured output. Each model must emit strict JSON containing one legal intent and a list of typed slot spans. Metrics are intent accuracy, intent macro-F1, slot micro-F1, exact match, and strict JSON-valid rate. Exact match requires the complete structured prediction to satisfy the evaluator.

4 Experimental Design

The primary matrix uses Gemma 4 E4B with 4-bit QLoRA. Every trained condition uses 500 optimizer steps and the same prompt template without a label catalog. The deployment-oriented zero-shot baseline uses a separate prompt containing the legal label catalog; it is not a same-prompt causal ablation. All headline evaluations use the untouched 2,974-row MASSIVE zh-TW test set.

The seed-42 matrix compares real-only training, conventional augmentation, unfiltered synthetic data at full and equal size, the filtered corpus, and the full-real upper reference. The primary paired analysis then trains real-only and real-plus-filtered conditions with seeds 42, 43, and 44 under identical data and configuration hashes. A 5,000-repetition hierarchical paired bootstrap first resamples seeds and then paired test rows within each selected seed. Exact McNemar tests are computed per seed for intent accuracy and exact match, with Holm correction over six tests.

Before observing the second-family results, the cross-family criterion was fixed: the paired mean delta for intent accuracy and exact match had to be positive in both families, and every corresponding hierarchical 95% confidence-interval lower bound had to exceed zero. Results are summarized per family and are not pooled.

5 Results

Table 1 reports the frozen seed-42 matrix. The filtered condition uses fewer synthetic rows than the full unfiltered condition but has the highest intent accuracy, slot F1, exact match, and JSON-valid rate among the synthetic conditions in this run.

Condition	Intent acc.	Macro-F1	Slot F1	Exact	JSON valid
Zero shot	10.66	23.12	0.00	8.10	17.38
Real only	73.54	75.20	62.14	49.06	98.02
Classical augmentation	74.31	75.59	62.58	46.81	96.23
Unfiltered full	75.99	76.42	65.01	51.21	97.75
Unfiltered equal-N	76.03	75.59	64.37	51.01	97.95
Filtered synthetic	76.19	76.09	66.54	52.12	97.98
Full real reference	84.53	81.65	71.58	60.66	99.73

Table 1: Seed-42 test metrics in percent.

Across the three paired Gemma seeds, filtered synthetic data improves intent accuracy by 4.14 ± 1.39 percentage points and exact match by 3.86 ± 0.73 points (mean $\pm$ sample standard deviation). The hierarchical bootstrap estimates are $+4.14$ with 95% CI $[2.60, 5.59]$ for intent accuracy and $+3.86$ with CI $[2.75, 4.92]$ for exact match. The six exact McNemar tests for these two metrics reject after Holm correction, with adjusted $p \leq 0.00017$. These tests apply only to the frozen rows and contract.

6 Robustness

The evaluation-only robustness manifest deterministically perturbs the test set with colloquial/lexical variants and ASR-like noise, yielding 8,922 rows per adapter. It never re-enters training. Across seeds 42–44, all five paired mean deltas are positive for both families. For Gemma, the filtered-minus-real-only deltas are 3.63 intent-accuracy points, 2.11 macro-F1 points,

2.75 slot-F1 points, 3.58 exact-match points, and 1.49 JSON-valid points. For Phi-4-mini, the corresponding deltas are 6.22, 4.69, 3.73, 6.98, and 1.83 points.

The uncertainty is important. Gemma macro-F1, slot F1, and JSON-valid sample standard deviations are comparable to or larger than their means at $n = 3$. Two Gemma metrics that are negative at seed 42 become positive when all three seeds are included. The perturbations are deterministic probes, not a measured distribution of real speech-recognition errors; they support a bounded stress test rather than a general robustness claim.

7 Cross-Family Replication

Phi-4-mini repeats the real-only versus real-plus-filtered comparison using the same corpus, prompt template, 500 steps, seeds, evaluator, and 2,974 test rows. The Phi intent-accuracy mean delta is +5.09 percentage points with hierarchical 95% CI [1.83, 9.02], and the exact-match delta is +4.71 with CI [1.36, 7.59]. The corresponding Gemma estimates are +4.14 [2.60, 5.59] and +3.86 [2.75, 4.92]. Both preregistered primary metrics therefore pass the cross-family criterion.

This result is not uniformly positive for every metric and test. Gemma’s JSON-valid interval crosses zero, and one Phi seed-level exact-match McNemar test does not reject after correction. The conclusion is consequently limited to replication across two families under this one frozen dataset and training contract; it is not a claim about arbitrary models.

8 Equal-N Recipe Ablation

The M19 composition experiment fixes each condition at 1,176 real plus 2,246 synthetic rows. An equal-size control retains all four recipes; four other conditions each exclude one recipe. Every condition uses Gemma, seed 42, 500 steps, and the same 2,974-row test set. Exact-match difference from control and a threshold of 2.5 percentage points were fixed before execution.

Excluded recipe	Intent acc.	Exact match	Exact delta
None (control)	75.99	49.50	0.00
Paraphrase	74.14	50.00	+0.50
Slot substitution	77.14	51.51	+2.02
Noise/code-switch	73.47	48.76	-0.74
Hard negative	76.19	50.81	+1.31

Table 2: M19 seed-42 equal-N composition comparison, in percent/percentage points.

The four exact-match deltas were +0.50 +2.02 -0.74 and +1.31 percentage points; none reaches the threshold. This is a descriptive comparison with a single seed ($n = 1$). It neither shows that the recipes are useless nor identifies an independent recipe effect. In particular, this result does not support a recipe-level causal claim. No additional seed, new metric, or lower threshold was added after inspecting the result.

9 Limitations

The core evidence uses one translated benchmark locale, two student families, three training seeds, a fixed 500-step budget, and one teacher/judge setup. MASSIVE zh-TW is translated from English SLURP and may not cover the distribution of naturally occurring Taiwanese speech. Synthetic

examples can inherit teacher biases, local-knowledge gaps, and generation modes. The F7 random stratum contains only 50 rows; its observed 6.0% rejection rate has a wide Wilson interval and targeted strata are not an unbiased corpus estimate.

M19 is a single seed and is intentionally underpowered for small recipe effects. Per-intent estimates vary considerably across seeds, so extreme seed-42 changes must not be interpreted as stable intent-specific effects. Robustness probes are generated perturbations rather than sampled production errors. There is no end-user study, no wall-socket energy measurement, and no evidence for other languages or domains. The report is not peer reviewed.

10 Ethics and Licensing

The source code is MIT licensed. MASSIVE seed utterances and the released synthetic dataset are distributed with CC BY 4.0 attribution and provenance described in the data card. The local Qwen teacher, gpt-oss judge, and primary Gemma student have their own Apache-2.0 upstream terms; Phi-4-mini is MIT licensed. These upstream terms are not replaced by the repository license.

The public model artifacts are the seed-42 filtered LoRA adapters for Gemma and Phi-4-mini. They are governed by the applicable Apache-2.0 and MIT terms, respectively, and neither repository redistributes base-model weights. The dataset contains task utterances rather than personal user logs, but synthetic content can still encode stereotypes or unsafe assumptions; downstream users should audit performance and harms in their deployment domain.

11 Resource Accounting

Every model workload ran locally on one NVIDIA GeForce RTX 4090 24 GB, with no paid model API usage. The frozen primary path accounts for 14.440 GPU hours. Including generation audits, extra seeds, robustness, Phi replication, and M19, the traceable local total is 42.412 hours.

At the documented 450 W GPU TDP, the total corresponds to a 19.085 kWh GPU-only upper-bound envelope. This value is not a wall-socket measurement and does not imply that the GPU operated at TDP throughout. One interrupted M19 attempt contributes 2.084 hours through a hash-anchored runtime audit so that discarded compute is not silently omitted.

12 Reproducibility

The public release corpus has SHA-256 `c65d7209d953e144299625f6a9224b98557b2677d55258a463a2992e5acf4665`. The published Gemma adapter is 155,609,536 bytes with SHA-256 `f70f423814dcd47943c92c0beb8b08a4e7f65e60a44355d3dcd95bed9f0bd60a`. The published Phi adapter is 92,309,112 bytes with SHA-256 `e9e4c77d79eb12da8cba64a7a484d260f753d000396752f51159e3c0f4e34376` and uses the fixed base revision `cfbefacb99257ffa30c83adab238a50856ac3083`. Machine-readable reports record row counts, data and prediction hashes, fixed seeds, model revisions, bootstrap settings, and interpretation boundaries.

The versioned source, this report, checksums, and evidence files are preserved through the GitHub–Zenodo integration under the all-versions DOI [10.5281/zenodo.21767492](https://doi.org/10.5281/zenodo.21767492). The public Phi adapter is available at <https://huggingface.co/steven0226/phi-4-mini-formosanlu-lora>; the public Gemma adapter and Dataset are linked from the source repository.

Repository checks rebuild reported README values from tracked reports, audit the sole-contributor history, verify documented commands, and validate publication metadata. The source reports are authoritative if prose and machine-readable evidence differ. Reproduction should use

the tagged environment and documented commands rather than rerunning generation with newer upstream model revisions.

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